

Lab 2: Binary Classification from Scratch

Sigmoid Classifier vs. Linear SVM

COMP 395 – Deep Learning

September 28, 2026

Task and methods

This experiment classifies the 569 samples in scikit-learn’s breast cancer dataset using 30 numerical features (0 = malignant, 1 = benign). The provided `load_data()` function uses an 80/20 train/test split with `random_state=42`, giving 455 training samples and 114 test samples. Each feature is standardized using only the training mean and standard deviation; the same transformation is applied to the test set.

The from-scratch model is one sigmoid neuron, $\hat{y} = \sigma(\mathbf{w}^\top \mathbf{x} + b)$, with loss $L = \frac{1}{2}(\hat{y} - y)^2$. Its gradients are computed explicitly:

$$\delta = (\hat{y} - y)\hat{y}(1 - \hat{y}), \quad \nabla_{\mathbf{w}} L = \delta \mathbf{x}, \quad \frac{\partial L}{\partial b} = \delta.$$

Weights and bias start at zero and are updated by per-sample stochastic gradient descent with learning rate 0.01 for 100 epochs, in fixed sample order. Predictions use a threshold of 0.5. The comparison model is `SVC(kernel="linear", random_state=42)`, with the default $C = 1$, trained on the same standardized training data. A linear SVM was retained to compare a margin-based objective with squared-error training while both models use linear decision boundaries.

Results

The sigmoid model’s average training loss decreased from 0.054478 in the first epoch to 0.007863 in the final epoch (Figure 1). These epoch losses average the losses observed before each sample’s update. Both models classified 449 of 455 training samples correctly. The sigmoid classifier correctly classified 113 of 114 test samples; the SVM correctly classified 109 (Table 1).

Table 1: Accuracy on the same fixed train/test split.

Model	Training accuracy	Test accuracy
From-scratch sigmoid classifier	98.68%	99.12%
Linear SVM ($C = 1$)	98.68%	95.61%

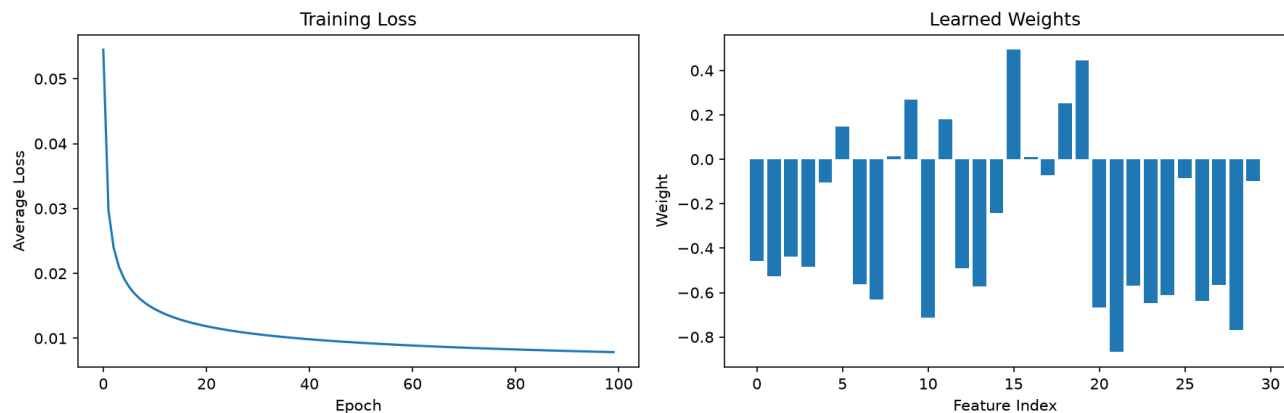


Figure 1: Average training loss and final feature weights of the from-scratch sigmoid classifier, generated by `binary_classification.py`.

Discussion

The from-scratch classifier performed better on this test split by 3.51 percentage points, or four additional correct predictions. Both models learn linear boundaries, but half-squared error and the SVM's margin-based objective with regularization can select different boundaries; this is a plausible explanation for the observed difference. It is not evidence that the from-scratch method is generally superior: there are only 114 test samples, no hyperparameter search was performed, and no repeated-split or cross-validation results were collected. The measured accuracies exceed the lab's training and test targets of 95% and 93%, respectively, for the from-scratch model.